Disasters and Business Formation: Evidence from a State-Month Panel of U.S. Business Applications

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Abstract

New business formation is a central driver of regional economic dynamism, yet little systematic evidence exists on whether natural disasters affect the rate at which new businesses are created, as distinct from their well-documented effects on aggregate output and employment. This paper examines the association between federally declared disasters and monthly business applications across U.S. states between July 2004 and June 2026, combining Business Formation Statistics from the U.S. Census Bureau with Disaster Declarations Summaries from the Federal Emergency Management Agency into a state-month panel of 13,464 observations across 51 states. A simple pooled correlation between disaster counts and business applications is weakly positive ($r \approx 0.11$). Once state fixed effects and month-year fixed effects are introduced to account for persistent regional differences and nationwide temporal shocks, the estimated association becomes negative and is not statistically distinguishable from zero at conventional significance levels ($\beta = -110.97$, clustered SE = 70.14, $p = 0.114$, 95% CI [-248.47, 26.52]). This pattern suggests that the weak positive correlation observed in the raw data more plausibly reflects the fact that larger, more disaster-prone states also generate more business applications overall, rather than a systematic within-state relationship between disaster timing and new business formation. The wide confidence interval indicates that this analysis lacks the precision to rule out economically meaningful effects in either direction, and the result should be read as an absence of detectable evidence under the present specification rather than evidence that no relationship exists. The paper contributes a reproducible, openly documented state-month panel linking two federal administrative data sources, a transparent empirical benchmark for a question examined to date primarily through case studies, and a fully reproducible computational pipeline with an accompanying interactive dashboard. The findings motivate future work incorporating lagged and severity-weighted disaster measures, finer geographic resolution, and more targeted causal identification strategies.

Index Terms*—*business formation, disaster declarations, two-way fixed effects, panel data, entrepreneurship, natural disasters, FEMA, regional economic resilience

I. INTRODUCTION

New business formation is widely regarded as a central driver of economic dynamism at both the regional and national level. Young firms disproportionately contribute to net job creation relative to their share of total employment [1], and the ongoing entry of new businesses into local economies is closely associated with innovation, the reallocation of resources toward more productive uses, and the capacity of a regional economy to adapt to changing conditions over time. For this reason, understanding what drives variation in the rate of new business formation across regions and over time has remained a persistent concern in entrepreneurship research, regional economics, and public policy alike. A substantial literature has documented that this variation reflects durable, region-specific characteristics — local industry structure, human capital, access to finance, and the broader entrepreneurial ecosystem in which new ventures are formed — that evolve slowly and differ markedly from one part of the country to another [2], [3].

Natural disasters represent a different, considerably more abrupt kind of economic disturbance to these regional economies, and one whose frequency and cost have drawn increasing attention from researchers and policymakers alike. The economic consequences of disasters have been studied extensively through their effects on aggregate output, employment, capital stock, and the pace of post-disaster recovery [4]–[6]. Considerably less is known, by comparison, about whether disasters influence the creation of entirely new businesses, as distinct from the survival or performance of firms that already exist at the time a disaster occurs. This distinction matters: a new business represents a fresh allocation of entrepreneurial effort and capital in response to a changed local environment, and its formation may respond to a disaster quite differently than the continued operation of an established firm would. Whether disasters are followed by more, fewer, or no significant change in new business activity remains an open empirical question, and one this paper is designed to address directly.

The reason this question does not have an obvious answer follows from the fact that disasters could plausibly influence entrepreneurial activity through several distinct, and in some cases directly competing, mechanisms. On one hand, a disaster may create new entrepreneurial opportunities by generating localized demand for reconstruction, contracting, and related recovery services, allowing individuals who recognize these emerging needs to found new ventures oriented specifically around them — a process consistent with the broader entrepreneurship literature on opportunity recognition [7]. Disruption to existing employment may also push displaced workers toward necessity entrepreneurship, founding businesses not because an attractive market opportunity has presented itself but because viable wage employment is no longer available [8]. On the other hand, a disaster may just as plausibly suppress new business formation in the short run, by damaging local infrastructure, disrupting consumer demand, and creating exactly the kind of uncertainty that discourages entrepreneurial risk-taking rather than encouraging it [9], [10]. A disaster may also alter a region's entrepreneurial ecosystem more durably, through population displacement or shifts in local capital availability, with consequences that could unfold over a considerably longer horizon than the disaster event itself. Because several of these mechanisms plausibly operate simultaneously and in offsetting directions, the net effect of a disaster on aggregate business formation is, in principle, ambiguous, and cannot be resolved by theoretical reasoning alone.

The existing empirical literature offers only partial guidance in resolving this ambiguity. Entrepreneurship research has established a great deal about why business formation varies across regions in ordinary times, and has recently gained access, through the Census Bureau's Business Formation Statistics program, to a genuinely timely national measure of new business applications at monthly frequency [11]. Separately, the disaster-economics literature has established that natural disasters carry substantial and heterogeneous economic consequences for regional economies, though this literature has concentrated primarily on aggregate output, employment, and recovery rather than on entrepreneurship specifically. The smaller body of work situated at the direct intersection of these two areas remains fragmented, consisting largely of case studies and conceptual accounts of entrepreneurship following individual disaster events [12], [13]. This work has usefully identified the mechanisms described above, but, to the author's knowledge, relatively little evidence exists on whether these mechanisms aggregate into a systematic, measurable pattern in business formation when examined across many states and many disasters using a consistent national panel, rather than within the context of a single closely studied event.

This paper addresses that gap directly. It asks whether the occurrence of federally declared natural disasters systematically influences state-level business applications in the United States, after accounting for persistent differences across states and shocks common to the entire country in a given month. This question is of interest to entrepreneurship researchers seeking a national, systematic benchmark against which more localized findings can be compared; to disaster economists concerned with the fuller range of economic consequences that follow a disaster beyond output and employment; and to regional policymakers responsible for designing post-disaster recovery strategies, for whom the question of whether disasters tend to accelerate or suppress new business creation carries direct practical relevance.

To answer this question, this study combines two publicly available federal administrative datasets: Business Formation Statistics from the U.S. Census Bureau and Disaster Declarations Summaries from the Federal Emergency Management Agency. These are merged into a monthly panel covering the fifty U.S. states and the District of Columbia, in which each state-month observation records the number of new business applications filed and the number of distinct federal disaster declarations affecting that state in that month. The empirical strategy centers on a two-way fixed effects regression, which includes state fixed effects to absorb persistent differences across states and month-year fixed effects to absorb shocks common to all states in a given month, with standard errors clustered at the state level to account for serial correlation in state-level disturbances over time. This design is intended to isolate the within-state, within-month association between disaster occurrence and business formation, net of two specific and economically important sources of confounding, though, as discussed at length later in this paper, it does not by itself establish a causal relationship between the two.

The descriptive relationship between disaster occurrence and business applications in the raw data is weakly positive, though modest in magnitude. Once state and month-year fixed effects are introduced, however, the estimated association becomes negative, and it is not statistically distinguishable from zero at conventional significance levels. This pattern suggests that the weak positive correlation observed in the raw data is more plausibly attributable to the fact that larger, more disaster-prone states also tend to generate more business applications overall, rather than to any systematic within-state relationship between the timing of disasters and the timing of new business formation. Taken together, the findings indicate that no systematic contemporaneous association between federally declared disasters and business applications is detectable under the empirical specification developed in this paper, a conclusion that is discussed at length, alongside its limitations and its implications for future research, in the sections that follow.

This paper makes several contributions. First, it constructs and documents a reproducible state-month panel linking two federal administrative data sources that, to the author's knowledge, have not previously been combined in this way, providing a national-level empirical resource for studying the relationship between disaster occurrence and entrepreneurial activity. Second, it offers a systematic empirical benchmark for a question that has, to date, been examined primarily through individual case studies and conceptual work, situating that question within a transparent panel data framework capable of separating cross-state from within-state variation. Third, it demonstrates concretely why this distinction matters, by showing that the weak positive association apparent in a simple pooled correlation weakens and reverses sign once state and time confounding are addressed — an empirical illustration with relevance well beyond this particular application. Fourth, the paper reports its central estimate honestly, including the fact that it is not statistically significant, rather than treating that outcome as a reason to search for an alternative specification, in keeping with a broader and increasingly emphasized commitment to transparent reporting in empirical social science. Finally, the analysis is accompanied by a fully documented, reproducible computational pipeline and an interactive dashboard that allow the underlying data, code, and results to be inspected and extended directly. None of these contributions is offered as a claim to have resolved the causal relationship between disasters and entrepreneurship; rather, they are offered as a carefully specified and transparently reported step toward a question that, as the discussion in later sections makes clear, will likely require more granular data and more targeted identification strategies to answer with greater confidence.

The remainder of this paper is organized as follows. Section II reviews the relevant literature in entrepreneurship, business formation, and disaster economics, and situates this study's contribution within it. Section III describes the two datasets used in this analysis and the construction of the state-month panel. Section IV presents the empirical strategy, including the two-way fixed effects specification and the reasoning behind the choice of clustered standard errors. Section V reports the descriptive and regression results. Section VI discusses the interpretation of these findings, with particular attention to reconciling the raw descriptive correlation with the fixed-effects estimate. Section VII documents the reproducibility of the analysis, including the accompanying computational pipeline and dashboard. Section VIII discusses the limitations of the study, and Section IX concludes and outlines directions for future research.

II. RELATED LITERATURE

A. Entrepreneurship and Business Formation

Research on the determinants of new business formation has a long history in entrepreneurship and regional economics, dating at least to Birch's [14] influential identification of small and young firms as a disproportionate source of net job creation in the United States. Subsequent work shifted attention toward explaining why rates of new firm formation vary so substantially across regions within a single national economy. Reynolds et al. [15] documented persistent cross-national and cross-regional variation in new firm formation rates, arguing that this variation reflects a combination of local market opportunity and the regional stock of entrepreneurial capability. Armington and Acs [2] extended this inquiry to U.S. regional variation specifically, showing that measures of local human capital, industry structure, and agglomeration economies help explain why some regions consistently generate more new business activity than others.

This literature has increasingly been reframed through the lens of entrepreneurial ecosystems — the idea that regional entrepreneurship reflects not simply aggregate economic conditions but a durable, locally embedded configuration of institutions, finance, culture, and support networks that shapes entrepreneurial activity over long periods [16], [17]. Feldman [3] similarly argued that firm formation should be understood as an inherently regional event, embedded in local knowledge spillovers, institutional resources, and networks that accumulate slowly and change little from one year to the next. This body of work establishes that new business formation is shaped by persistent, region-specific characteristics that are unlikely to shift materially within the time horizon of a single study, a point directly relevant to the present study's inclusion of state fixed effects, which are designed to absorb precisely this kind of slow-moving regional heterogeneity — entrepreneurial ecosystem strength among them — rather than to explain it.

Business Formation Statistics, the primary outcome measure used in this paper, are a comparatively recent addition to this literature. The series was developed by the U.S. Census Bureau to provide a more timely proxy for entrepreneurial activity than existing administrative measures of firm entry, which typically lag actual business creation by a year or more. Bayard et al. [11] describe the construction of this series from Employer Identification Number applications and show that it tracks subsequent, more comprehensive measures of business formation closely while being available at monthly frequency. This timeliness is central to why BFS was chosen as the outcome measure here: a research question concerned with the state-month timing of disasters relative to business formation requires an outcome series with genuine monthly resolution, which most alternative administrative measures of firm entry do not provide. Haltiwanger et al. [1] further show that young and small firms play an outsized role in aggregate job creation and reallocation relative to their share of total employment, reinforcing the broader motivation for studying the conditions, including disruptive local events, under which such firms come into existence in the first place.

B. Economic Effects of Natural Disasters

A separate and considerably larger literature examines the economic consequences of natural disasters at the regional level, typically with greater emphasis on aggregate output, employment, and long-run growth than on entrepreneurship specifically. This literature reflects a persistent tension between two competing views of how disasters affect local economies: one emphasizing destruction, in which damaged capital and interrupted production depress local economic activity, at least in the short run; and one emphasizing reconstruction, in which post-disaster investment in rebuilding can generate offsetting, or even net-positive, economic activity.

Skidmore and Toya [4] offered an early and influential treatment of this tension, finding that a higher frequency of climatic disasters is associated with higher long-run total factor productivity growth across countries, consistent with disasters accelerating the replacement of older capital stock. This finding has not been uniformly replicated at more localized levels of geography. Strobl [5], examining hurricane strikes across U.S. coastal counties, found that hurricanes reduce county-level economic growth in the short run, with effects concentrated in the years immediately following a storm. Belasen and Polachek [18], studying Florida county labor markets following hurricane strikes, found a more nuanced pattern: worker earnings in directly affected counties rose in the short run relative to unaffected counties, while employment growth slowed, consistent with a localized labor demand shock rather than a uniform economic contraction. Cavallo and Noy [19], surveying this literature broadly, concluded that the economic effects of disasters are highly heterogeneous across studies and sensitive to the level of geographic aggregation, the type of hazard considered, and the time horizon examined, with more localized analyses more likely than coarser ones to detect negative short-run effects.

Work on recovery dynamics and regional resilience has further emphasized that disaster impacts evolve over time in ways a purely contemporaneous analysis can miss. Deryugina [20], studying the fiscal and economic aftermath of Hurricane Katrina, found that federal disaster aid and social insurance programs shaped the economic trajectory of affected areas over several years, with effects not fully apparent in the immediate aftermath. Hsiang and Jina [6], using a global sample of tropical cyclones, found that national output growth remains depressed for as long as a decade following a severe cyclone strike. Collectively, this literature suggests that any economic response to disaster, destructive or reconstructive, is unlikely to be confined to the month, or even the year, in which the disaster occurs — a point directly relevant to the contemporaneous specification adopted in this study and revisited as a limitation in Section VIII.

C. Disasters and Entrepreneurship

The literature examining entrepreneurship specifically in the aftermath of disaster is considerably smaller than the broader disaster-economics literature summarized above, and its findings are correspondingly less settled. This smaller literature has generally combined case-based and conceptual work with a limited number of quantitative studies of business survival and formation in affected areas, and it offers a useful entrepreneurship-specific lens on the destruction-versus-reconstruction tension already noted in Section II-B.

Some work in this space emphasizes conditions under which disasters can give rise to new entrepreneurial activity rather than merely suppressing it. Williams and Shepherd [12], studying venture creation following the 2010 Haiti earthquake, found that some individuals responded to the disruption by founding new ventures explicitly oriented toward addressing needs created by the disaster itself, framing this as a form of opportunity recognition — the process by which individuals identify and act on previously unexploited possibilities for value creation [7] — that is activated rather than foreclosed by extreme adversity. Monllor and Murphy [13] develop a conceptual account of disaster entrepreneurship along similar lines, arguing that the destruction of existing economic activity can simultaneously create openings for new venture creation oriented around recovery and rebuilding needs. This work connects naturally to the long-standing distinction in the entrepreneurship literature between opportunity entrepreneurship, motivated by the pursuit of a perceived market opportunity, and necessity entrepreneurship, in which individuals turn to self-employment for lack of viable alternative employment [8]. A disaster plausibly stimulates both forms simultaneously: rebuilding demand may create genuine opportunities in construction and related trades, while job losses in disrupted sectors may push displaced workers toward necessity-driven self-employment, and a simple aggregate count of new business applications cannot distinguish between these two, potentially offsetting, mechanisms.

Other work emphasizes the substantial barriers disasters place in the way of business formation and survival. Runyan [9], studying small businesses following a natural disaster, identified damaged infrastructure, disrupted local demand, and difficulty accessing credit and insurance payouts as central barriers to recovery, a set of obstacles that would be expected to suppress rather than stimulate new business activity in the short run. More broadly, the entrepreneurship literature on resilience and entrepreneurial action under conditions of extreme uncertainty offers a further reason to expect ambiguous or context-dependent effects. McMullen and Shepherd [10], building on Knight's [21] classical distinction between measurable risk and true uncertainty, argue that entrepreneurial action itself is shaped by how individuals perceive and tolerate uncertainty, a perception plausibly altered, in either direction, by exposure to a disaster. Bullough et al. [22], studying entrepreneurial intentions in conditions of extreme adversity, similarly find that individual resilience and self-efficacy play a central role in whether adverse and uncertain conditions translate into entrepreneurial action or withdrawal from it.

This disagreement across studies is unlikely to reflect simple inconsistency in the underlying phenomenon so much as genuine heterogeneity in mechanisms, combined with substantial differences in research design. Disaster severity varies enormously across the events different studies examine, as does the industrial composition of affected areas, the generosity of recovery assistance, individual entrepreneurs' resilience and risk tolerance, and the level of geographic and temporal aggregation used to measure any resulting change in entrepreneurial activity. A study examining new venture creation following a single catastrophic event in one community is answering a different empirical question than one pooling many disasters of varying severity across an entire country over two decades, and there is little reason to expect the two to yield identical conclusions. This heterogeneity suggests that the relationship between disasters and entrepreneurship, if it exists in any general form, is likely conditional on severity, local context, and individual-level factors that a simple state-month administrative disaster count is poorly positioned to capture — a concern that directly motivated the interpretation offered in Section VI.

D. Panel Data and Fixed Effects in Regional Economics

Fixed effects estimators are a standard tool in regional economics for addressing confounding that a purely cross-sectional comparison cannot resolve [23], [24]. Region fixed effects absorb persistent, unobserved differences in geography, industrial structure, and entrepreneurial ecosystem strength that would otherwise confound a comparison across regions, while time fixed effects absorb shocks common to all regions in a given period, such as national recessions or, in the present setting, the nationwide disruption associated with the COVID-19 pandemic. Combining both, as in the two-way fixed effects design used here, restricts identification to variation within a given state over time, net of whatever is common across states in a given month.

Recent econometric work has clarified important limitations of this approach, particularly in settings involving staggered treatment timing or effects that vary across units and over time. Goodman-Bacon [25] and de Chaisemartin and D'Haultfœuille [26] show that in difference-in-differences settings with staggered adoption, standard two-way fixed effects estimates can implicitly weight comparisons in ways that bias the estimated coefficient even absent confounding of the traditional kind. The present setting differs from the canonical staggered-adoption case, since disaster count is a continuously varying, repeatedly observed measure rather than a single binary treatment, but this literature is a useful reminder that a two-way fixed effects specification does not automatically guarantee an unbiased or straightforwardly causal estimate. This reinforces the caution expressed in Sections IV through VI: the specification used here addresses two specific, well-understood sources of confounding, and no more.

E. Research Gap

Taken together, this literature leaves a reasonably well-defined gap that the present study is positioned to address, without claiming to resolve any single element of it for the first time. The entrepreneurship literature has established that business formation is shaped by persistent regional and ecosystem-level characteristics and has recently gained access, through Business Formation Statistics, to a genuinely timely national measure of formation at monthly frequency. The disaster-economics literature has established that natural disasters carry substantial and heterogeneous economic consequences that typically unfold over a longer horizon than a single month. The smaller literature at the direct intersection of these two areas remains fragmented across conceptual work and individual case studies, pointing toward opportunity recognition and necessity-driven self-employment as plausible, potentially offsetting mechanisms, without offering a systematic, national-level empirical benchmark against which these mechanisms can be assessed in aggregate.

What remains genuinely unresolved is not whether disasters could plausibly reshape entrepreneurial behavior — the conceptual and case-based literature reviewed above offers good reason to think they might — but whether the several mechanisms this literature identifies actually leave a detectable footprint in aggregate business formation once measured systematically, across many states and many disasters of varying character, rather than within a single community or event. A disaster is, in principle, capable of activating opportunity recognition among individuals who identify unmet local needs created by recovery and rebuilding; of pushing displaced workers toward necessity entrepreneurship in the absence of viable wage employment; of catalyzing recovery-driven venture creation concentrated in construction, contracting, and related trades; and of altering the local entrepreneurial ecosystem more durably, through shifts in population, capital availability, or institutional attention that persist well beyond the disaster itself. These mechanisms are not mutually exclusive, and several plausibly operate at once, in offsetting directions, within the same affected state and the same narrow window of time. Existing work has documented that each of these mechanisms can occur in specific, closely studied settings, but it has not established whether they aggregate into a measurable, generalizable association at the level of a state economy in a given month, once the substantial regional heterogeneity in baseline entrepreneurial activity documented in Section II-A is properly accounted for.

This study contributes to that gap by combining two federal administrative data sources — Census Bureau Business Formation Statistics and FEMA Disaster Declarations — that, to this author's knowledge, have not previously been linked into a systematic, national, state-month panel for the study of entrepreneurial activity, and by applying a transparent two-way fixed effects specification explicitly designed to isolate the within-state, within-month variation that any of the mechanisms above would need to produce in order to be visible at this level of aggregation. In reporting an estimated association that is not statistically distinguishable from zero, this study does not settle the conceptual question of whether opportunity recognition, necessity entrepreneurship, or recovery-driven venture creation operate in the aftermath of disaster; rather, it provides evidence that, whatever combination of these mechanisms is at work, their net effect is not detectable as a systematic, national-level pattern in monthly business formation using an administrative measure of disaster occurrence. That distinction is itself a meaningful empirical contribution, and it points toward the more granular, mechanism-specific research designs outlined in Section IX as the necessary next step for determining which, if any, of these entrepreneurial responses to disaster dominate in practice.

III. DATA

A. Business Formation Statistics (BFS)

The dependent variable in this study is drawn from the Business Formation Statistics (BFS) program administered by the U.S. Census Bureau [27], a monthly series constructed from Employer Identification Number (EIN) applications filed with the Internal Revenue Service. The BFS program was designed to provide a timelier proxy for entrepreneurial activity than annual business registry data, and it has become a standard source for measuring new business formation in the United States at sub-national levels of geography.

The BFS release used in this study reports several related series, distinguished by whether the underlying application shows characteristics associated with an eventual employer business, and by whether counts are seasonally adjusted. This analysis uses the headline Business Applications series (`BA_BA`), which captures the full population of new EIN applications rather than a filtered subset conditioned on projected employment status. The not-seasonally-adjusted variant of this series is used throughout, a deliberate choice explained below.

BFS data are published at multiple levels of geographic and industry aggregation. At the national level, the series is further disaggregated by two-digit NAICS industry sector; at the state level, however, only the aggregate total across all industries is available. Because the empirical analysis in this paper is conducted at the state-month level, the industry-disaggregated series is not usable here, and the analysis instead relies on the state-level total (`naics_sector = TOTAL`). The sample is restricted to the fifty U.S. states and the District of Columbia; Puerto Rico is present in the Business Applications series but is excluded from the Business Formation (realized-formation) series entirely, and is dropped from this analysis to maintain a geographically consistent panel.

The raw BFS release used in this study reports monthly application counts in a wide format, with one column per calendar month and one row per combination of seasonal-adjustment status, industry sector, series type, geography, and year. Three non-numeric disclosure codes appear within the monthly count columns, following standard Census Bureau disclosure-avoidance conventions: a value is coded as suppressed when reporting it would risk disclosing information about individual establishments, as failing to meet Census Bureau publication-quality standards, or as not available. These codes are treated as distinct from ordinary missing values throughout the construction of the analysis panel, since each reflects a different substantive circumstance rather than a simple gap in the source data.

Because the not-seasonally-adjusted series is used, monthly counts retain their natural calendar-month variation rather than having that variation statistically removed prior to release. This choice is made deliberately: seasonal adjustment is designed to remove exactly the kind of month-to-month variation that this study's identification strategy relies on to detect an association between disaster timing and business formation. Working with the unadjusted series, combined with the month-year fixed effects described in Section IV, allows the empirical model to absorb common seasonal and macroeconomic patterns directly from the data rather than relying on the Census Bureau's own seasonal-adjustment procedure, which is estimated independently of this study's disaster measure.

B. FEMA Disaster Declarations

The independent variable of interest is constructed from the Disaster Declarations Summaries dataset maintained by the Federal Emergency Management Agency (FEMA) through its OpenFEMA platform [28]. This dataset records every federal disaster declaration since 1953, including major disaster declarations, emergency declarations, and fire management declarations, together with the incident type, the geographic area affected, and the relevant dates associated with each declaration.

Two aspects of the raw FEMA data required deliberate handling before it could be merged with the BFS panel. First, the raw dataset reports one row per county (or, in some cases, per other sub-state designated area, such as a tribal reservation) affected by a given disaster, rather than one row per disaster. A single federally declared disaster event can therefore generate dozens of rows in the raw file. Direct verification of the underlying data confirmed that each disaster declaration, identified by its `disasterNumber`, is associated with exactly one state; the row-level multiplicity in the raw file arises entirely from sub-state geographic detail, not from any disaster spanning multiple states under a shared declaration number. This made it possible to collapse the raw county-level records to one row per disaster-state pair by de-duplicating on the combination of `disasterNumber` and state, without needing to apportion a single event across several states. This collapsing step reduced the raw file from 70,049 county-level records to 5,027 distinct state-level disaster events.

Second, the FEMA dataset reports several dates associated with each declaration, including the date of the underlying incident and the date on which the declaration was formally approved. This study uses the incident begin date (`incidentBeginDate`), rather than the declaration date, as the timing anchor for each event. The declaration date reflects the completion of an administrative review process and can lag the onset of the physical event by a period ranging from days to several weeks; anchoring the analysis to the incident date instead keeps the disaster measure aligned with the timing of the underlying physical event rather than with the pace of the federal approval process.

The FEMA dataset also includes several U.S. territories and freely associated states not present in the BFS geography (American Samoa, the Federated States of Micronesia, Guam, the Marshall Islands, the Northern Mariana Islands, Palau, and the U.S. Virgin Islands). These are excluded from the analysis, consistent with the geographic scope established for the BFS side of the panel.

C. Construction of the Analysis Panel

The analysis panel is constructed by aggregating the cleaned FEMA event data to a state-month disaster count, and merging this count onto the cleaned BFS state-month panel. The disaster count for a given state-month is defined as the number of distinct disaster declarations, by `disasterNumber`, whose incident begin date falls within that calendar month for that state.

The BFS side of the panel serves as the structural backbone of the merge. Because BFS reports a continuous monthly series for every state from July 2004 onward, left-joining the sparse FEMA disaster panel onto the BFS panel guarantees that every state-month present in the Business Applications series is retained, whether or not a disaster was declared in that period. State-months with no matching disaster record are assigned a disaster count of zero rather than a missing value, reflecting the fact that the absence of a FEMA declaration in a given state-month is itself a meaningful and directly observed outcome, not an unobserved data gap. This distinction matters for the empirical analysis: treating no-disaster months as missing would discard the majority of the panel's variation and would misrepresent the absence of a declared disaster as a data limitation rather than as evidence.

The resulting merged panel spans 50 states and the District of Columbia (51 geographic units in total), covering 270 months from July 2004 through December 2026, for 13,770 state-month observations. Of these, 306 observations have a missing value for business applications, corresponding to the most recent months of 2026 for which the Census Bureau has not yet published data at the time this analysis was conducted; no other missing values are present in either the business applications or disaster count variables. These 306 observations are excluded from the descriptive and regression analyses reported in Section V, yielding an estimation sample of 13,464 state-month observations, still evenly distributed across all 51 states. No duplicate state-month keys were found at any stage of construction, and every FEMA disaster count value used in the merge was independently reconciled against the underlying event-level data prior to use.

One further reconciliation point is worth noting explicitly. Because FEMA's disaster declarations extend back to 1953, while the BFS monthly series begins only in July 2004, a portion of the FEMA record — 1,790 state-month disaster observations — falls entirely outside the period covered by BFS and is therefore not represented in the merged panel. This is an expected consequence of the differing historical coverage of the two source datasets rather than an artifact of the merge procedure, and it does not affect the internal consistency of the resulting panel over the period both datasets jointly cover.

D. Descriptive Statistics

Table I reports summary statistics for the two variables central to this analysis, computed over the 13,464-observation estimation sample. Monthly business applications average 5,715 per state, with a median of 3,282, indicating a right-skewed distribution consistent with the wide variation in population and economic activity across states. The minimum observed value is 237 and the maximum is 83,425, with a standard deviation of 7,854; the interquartile range spans from 1,285 to 6,435. Fig. 1 presents the distribution of this variable across all state-months, and Fig. 2 presents its distribution disaggregated by year, both of which confirm the right-skewed, heavy-tailed shape suggested by the summary statistics.

Statistic	Business Applications
Count	13,464
Mean	5,715.14
Std. Dev.	7,853.69
Min	237
25th pct.	1,285
Median	3,281.5
75th pct.	6,435.25
Max	83,425

TABLE I. SUMMARY STATISTICS, ESTIMATION SAMPLE

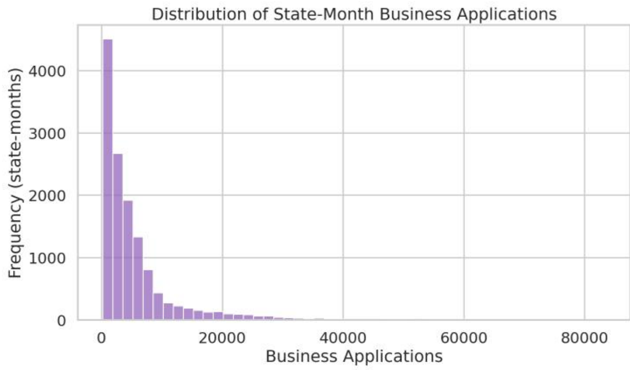

Fig. 1. Distribution of state-month business applications.

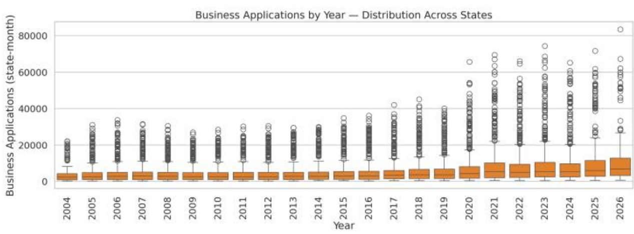

Fig. 2. Business applications by year, distribution across states.

A simple pairwise Pearson correlation between disaster count and business applications, reported in Table III below, is 0.106, indicating a weak positive linear association at the raw, unadjusted level.

The disaster count variable is considerably more concentrated at zero: 13.38 percent of state-months in the full merged panel have at least one declared disaster, with a maximum of 20 distinct declarations in a single state-month. Table II reports descriptive statistics for business applications disaggregated by whether a disaster was declared in that state-month. State-months with at least one disaster declaration show a higher mean (6,972) and higher standard deviation (9,911) than state-months without a disaster declaration (mean 5,516; standard deviation 7,457), while the medians of the two groups are considerably closer (3,402 versus 3,261). Fig. 3 presents this comparison as a boxplot, which shows substantial overlap between the two distributions alongside the difference in means reported in Table II.

Group	N
No disaster month	11,623
Disaster month	1,841

TABLE II. BUSINESS APPLICATIONS BY DISASTER STATUS

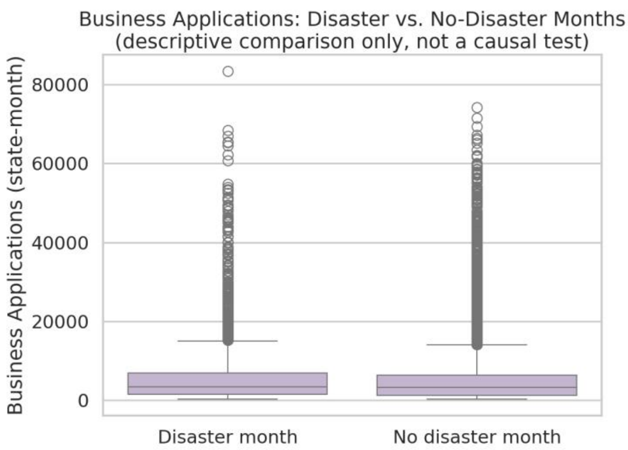

Fig. 3. Business applications: disaster vs. no-disaster months (descriptive comparison only).

As discussed in Section VI, this raw correlation should not be interpreted as evidence of a meaningful relationship between disaster occurrence and business formation in isolation, since it does not account for systematic differences between states that experience more disasters and states that experience fewer.

	Bus. Applications
Business Applications	1.000
Disaster Count	0.106
Year	0.230

TABLE III. PEARSON CORRELATION MATRIX

Table III also reports the correlation between business applications and calendar year (0.230), reflecting the general upward trend in business applications documented in Fig. 1, and the near-zero correlation between disaster count and calendar year (-0.013), indicating that the frequency of declared disasters in this sample does not exhibit a strong secular trend over the period studied. The empirical strategy described in Section IV is designed specifically to address the limitations of this unadjusted correlation by isolating within-state, within-month variation.

IV. EMPIRICAL STRATEGY

A. Model Specification

The relationship between disaster occurrence and new business formation is estimated using a two-way fixed effects (TWFE) panel regression, implemented exactly as specified in the accompanying analysis code. The estimating equation is:

$$Y_{it} = \beta \cdot D_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \quad (1)$$

where i indexes the 51 units of geography in the panel (the fifty U.S. states and the District of Columbia) and t indexes calendar month, running monthly from July 2004 through the most recent month for which Business Formation Statistics data are available. Here Yit denotes business applications, Dit denotes disaster count, αi are state fixed effects, γt are month-year fixed effects, and εit is the error term.

The dependent variable, business applications, is the count of new business applications filed in state i during month t, drawn from the not-seasonally-adjusted BFS series described in Section III-A. Monthly business applications are used as the outcome measure because they represent the most direct and timely observable proxy for new entrepreneurial activity available at the state level; unlike measures of business survival, revenue, or employment, an application count captures the initial decision to start a business without conflating it with the business's subsequent performance. This makes it a natural outcome variable for a study concerned with whether disaster exposure changes the rate at which new businesses are started, rather than how existing businesses fare afterward.

The independent variable of interest, disaster count, is the number of distinct federal disaster declarations with an incident begin date falling in state i during month t, as constructed in Section III-C. This is the only time-varying regressor in the model; no additional covariates are included.

The terms αi and γt are state and month-year fixed effects, respectively, each implemented as a full set of indicator variables absorbed directly by the `PanelOLS` estimator from the `linearmodels` package [29], rather than estimated as explicit dummy coefficients. The state fixed effects αi absorb any characteristic of a state that does not vary over the sample period, while the month-year fixed effects γt absorb any factor common to all states in a given calendar month. The error term εit captures remaining state-month-specific variation not explained by the disaster count or by either set of fixed effects.

The coefficient of primary interest is β, which measures the within-state association between an additional disaster declaration and the level of business applications in that same state-month, net of the two sets of fixed effects described above. This coefficient should be understood strictly as a measure of statistical association conditional on the included fixed effects, not as an estimate of the causal effect of disasters on business formation; the conditions under which such a causal interpretation would be warranted, and why they are not fully satisfied here, are discussed later in this section and revisited in the Discussion.

B. Identification Logic and Fixed Effects

The choice of a two-way fixed effects specification follows directly from a basic concern raised by the descriptive evidence in Section III-D: states that experience more federally declared disasters are not a random subset of U.S. states. They tend to differ systematically from states that experience fewer disasters in ways that are also likely to matter for business formation — in geography, climate exposure, population size, industrial composition, and the underlying pace of entrepreneurial activity that a state exhibits in an ordinary year. A regression of business applications on disaster counts that ignored these differences would risk attributing to disasters what is in fact simply a feature of which states tend to be disaster-prone in the first place.

The state fixed effects α_i are included specifically to remove this source of confounding. By allowing each state to have its own baseline level of business applications, the model absorbs any characteristic of a state that is fixed, or nearly fixed, over the sample period — its geography and climate exposure, the long-run structure of its economy, its population size, and any persistent regional entrepreneurial culture. Once these state fixed effects are included, the coefficient β is no longer estimated by comparing high-disaster states to low-disaster states; it is estimated only from month-to-month movement in disaster counts within the same state.

This within-state comparison, however, remains vulnerable to a second source of confounding: events that affect all states simultaneously in a given month. National recessions, seasonal patterns in business formation common across the country, changes in federal tax or regulatory policy, and broad economy-wide shocks — most notably the disruption associated with the COVID-19 pandemic and the subsequent surge in business applications documented in Section III-D — could easily be correlated with both nationwide disaster activity in a given month and nationwide business formation activity in that same month, independent of any relationship between the two at the state level. The month-year fixed effects γ_t are included to remove exactly this concern, by allowing each calendar month to have its own nationwide baseline level of business applications shared across all states.

Taken together, these two sets of fixed effects mean that β is identified purely from within-state deviations from a state's own average, occurring in months that also deviate from the nationwide average for that month. In practical terms, the model asks: in months when a given state saw an unusually high or low number of disaster declarations relative to its own typical pattern, and after netting out whatever was happening nationally that month, did that state's business applications also move in a corresponding direction? This is a considerably more demanding empirical question than a simple cross-sectional or pooled correlation, and it substantially reduces the scope for omitted-variable bias relative to the raw association reported in Section III-D.

It is important to be precise about what this design does and does not accomplish. Ruling out confounding from factors that are constant within a state, or common across all states in a given month, is not the same as ruling out all confounding. A state could still experience a disaster and an unrelated economic disturbance within the same month for reasons specific to that state and that month — a local industry shock coinciding with a severe weather event, for instance — and the fixed effects described above would not separate these two influences from one another. Reverse causality, in which conditions affecting business formation also affect the likelihood or administrative handling of a disaster declaration, is likewise not addressed by this design. For these reasons, this remains an observational study, and the coefficient β should be read as an estimate of association net of two specific and economically meaningful sources of confounding, not as a causal effect of disasters on entrepreneurial activity.

C. Inference and Clustered Standard Errors

Standard errors in this model are clustered at the state level, following standard practice for panel data in which the same cross-sectional unit is observed repeatedly over time. Shocks to business formation within a given state are unlikely to be independent across consecutive months: a state's local economic conditions, administrative processes, and business cycle position tend to persist and evolve gradually rather than resetting each month. Conventional standard errors, which assume independent and identically distributed errors, would treat each of a state's 264 monthly observations as providing independent information, overstating the effective sample size and understating the true uncertainty around β. Clustering by state allows the error terms for a given state to be arbitrarily correlated with one another across time, while assuming independence across different states, which is a considerably more realistic description of how disturbances to business formation are likely to behave in this setting.

Before turning to the coefficient estimate itself, it is useful to confirm that the fixed effects included in the model are statistically warranted rather than an arbitrary specification choice. The model output includes an F-test for poolability, which tests the joint null hypothesis that all state and month-year fixed effects are equal to zero — that is, that a simple pooled regression without any fixed effects would fit the data equally well. This test returns a statistic of 257.27 ($p < 0.001$), indicating that the fixed effects jointly contribute significant explanatory power relative to a pooled specification. This result supports the decision to include both sets of fixed effects rather than estimating a simpler pooled model, though it speaks only to the appropriateness of the fixed-effects structure itself and carries no implication, on its own, about the disaster count coefficient, which is discussed in Section V.

V. RESULTS

A. Descriptive Patterns Before Regression

Before turning to the fixed effects specification, it is useful to consider what the raw, unadjusted data suggest about the relationship between disaster declarations and new business formation. Monthly business applications across the estimation sample are highly variable, both across states and across time, with a distribution that is markedly right-skewed: a small number of state-months, corresponding to the largest and most economically active states, generate application counts many times the typical level observed elsewhere in the panel. The bulk of state-months cluster at comparatively modest levels, with the median well below the mean, a pattern consistent with the underlying reality that a handful of large states — California, Texas, and Florida prominent among them — account for a disproportionate share of national business formation activity in any given month.

Disaster declarations, by contrast, are a comparatively rare event at the state-month level. Only about one in eight state-months in the sample includes at least one federally declared disaster, and the large majority of state-months record no disaster activity at all. When business applications are compared across these two groups, state-months with a disaster declaration show a somewhat higher average level of business applications than state-months without one, and a wider spread around that average. The median values for the two groups, however, are considerably closer to one another than the means, suggesting that the difference in averages is driven disproportionately by a subset of high-activity state-months rather than reflecting a uniform shift across the distribution. The boxplot comparison in Fig. 3 makes this pattern visually clear: the interquartile ranges overlap substantially, and the apparent difference in central tendency is modest relative to the dispersion present within each group. A scatter plot of disaster counts against business applications (Fig. 4) tells a similar story — state-months with a larger number of declared disasters are not visibly concentrated toward higher or lower application counts, and the relationship, if any, is not detectable by eye against the scale of the underlying variation.

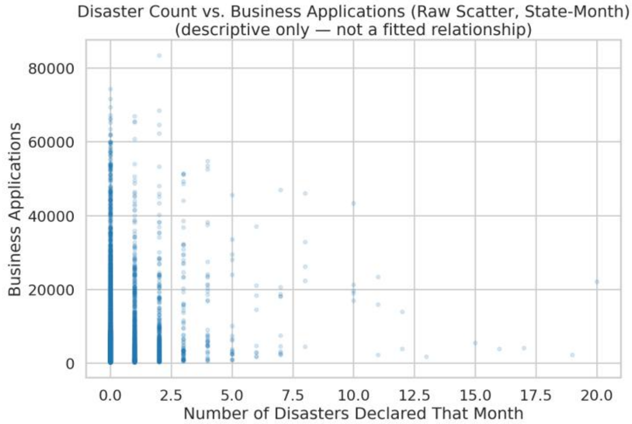

Fig. 4. Disaster count vs. business applications, raw scatter (descriptive only).

Formalizing this pattern, the Pearson correlation between disaster count and business applications across the full estimation sample is approximately 0.106 — positive, but weak by any conventional standard. Considered together with the descriptive comparisons above, this raw association is compatible with several very different underlying stories. It could reflect a genuine, if modest, tendency for business applications to rise in months when a disaster is declared. It could equally reflect nothing more than the fact that large, economically active states — which generate more business applications simply as a function of their size — also happen to accumulate more disaster declarations over time, for reasons entirely unrelated to any month-specific relationship between the two. The descriptive evidence on its own cannot distinguish between these possibilities. It establishes that some positive association exists in the raw data, and it motivates a closer empirical treatment, but it does not by itself provide a basis for drawing conclusions about entrepreneurial behavior following a disaster. Separating a genuine within-state relationship from a between-state artifact of this kind is precisely the task taken up by the fixed effects specification introduced in Section IV, and it is to that specification's results that the analysis now turns.

B. Main Regression Results

The two-way fixed effects regression yields a coefficient on disaster count of -110.97, with a state-clustered standard error of 70.14 (Table IV). This corresponds to a t-statistic of -1.58 and a p-value of 0.1136, and a 95 percent confidence interval spanning -248.47 to 26.52.

Variable	Coef.	SE	
disaster_count	-110.97	70.14	-1.

TABLE IV. TWO-WAY FIXED EFFECTS REGRESSION RESULTS

Substantively, the point estimate suggests that, within a given state and after removing whatever was happening nationally in that same month, an additional disaster declaration is associated with roughly 111 fewer business applications in that state-month. However, the confidence interval surrounding this estimate is wide, and it includes zero. This means that, at conventional significance levels, the estimated association is not statistically distinguishable from zero: the data are consistent with a true relationship that is negative, with one that is positive, or with no relationship at all, once the two sets of fixed effects described in Section IV have absorbed persistent state-level differences and nationwide monthly shocks.

This result is worth pausing on, because it stands in some contrast to the weak positive raw correlation described in Section V-A. Once the analysis moves from a simple pooled comparison to a within-state comparison over time — the empirical exercise the fixed effects specification is designed to perform — the sign of the estimated relationship reverses, and whatever association remains is no longer estimated with enough precision to rule out zero. This is consistent with the interpretation offered above: that the raw positive correlation is more plausibly a reflection of which states tend to experience more disasters generally, rather than evidence that disaster months themselves are followed by higher business formation within a given state. The regression results do not, however, allow this study to make the stronger claim that no relationship exists. A confidence interval this wide is compatible with economically meaningful effects in either direction — a genuine decline in business formation following a disaster, a genuine increase associated with post-disaster rebuilding activity, or effects close to zero — and the data at hand do not have the statistical power to adjudicate between these possibilities. What can be said, with appropriate caution, is that this analysis does not provide strong empirical evidence of a systematic short-run association between federally declared disasters and business application activity at the state-month level, once persistent state characteristics and common nationwide shocks are taken into account.

It is worth distinguishing explicitly between the statistical and economic dimensions of this result. Statistical significance speaks to whether an estimated relationship can be reliably distinguished from the possibility of no relationship, given the amount and structure of the data available; it does not, on its own, speak to whether an effect would matter in practice were it to exist. Here, both considerations point toward caution rather than a firm conclusion in either direction: the estimate is not statistically significant, and even its point estimate — a reduction of roughly 111 applications per state-month — is modest set against the average of several thousand monthly applications typical of the panel, though this magnitude comparison is offered only as context, since the estimate itself cannot be treated as a reliable characterization of the true relationship given its lack of statistical precision.

One further methodological point bears mention. The model output reports two versions of the overall F-statistic: a conventional version, which assumes independent and identically distributed errors, and a robust version computed under the clustered covariance structure described in Section IV-C. These two versions can and do diverge, precisely because they make different assumptions about how errors are correlated within a state over time. Because this study clusters standard errors by state to account for the plausible persistence of state-level shocks, inference throughout this paper is based on the clustered, robust statistics rather than the conventional ones; the conventional F-statistic is reported in the full regression output for completeness but is not treated as the basis for judging statistical significance in this analysis.

C. Model Diagnostics

The fixed effects specification explains very little of the total variation in business applications: the within-state R^2 is 0.0001, the between-state R^2 is -0.0052, and the overall R^2 is -0.0045 (Table V). Considered in isolation, these values are low by the standards of many applied settings, but this is not a distinctive weakness of this particular specification. A single time-varying regressor — disaster count — is being asked to explain month-to-month movements in business applications after everything structurally distinctive about a state, and everything common to a given month nationally, has already been removed. Business formation is a behavior shaped by a wide range of local economic conditions, most of which are not captured by disaster activity alone, and a low within-state R^2 in this setting reflects the modest share of month-to-month variation that any single variable of this kind could plausibly explain, rather than indicating that the model has been misspecified.

Metric	
N (observations)	
N (states)	
N (time periods)	
R^2 (within)	
R^2 (between)	
R^2 (overall)	
F-statistic (poolability)	2

TABLE V. MODEL DIAGNOSTICS

The distribution of model residuals departs from normality. A histogram of the residuals (Fig. 7) shows a sharp central concentration with pronounced tails extending in both directions, and the accompanying quantile-quantile plot (Fig. 8) shows the residuals diverging from the reference line particularly in the upper tail, consistent with the heavy-tailed, right-skewed character of the underlying business applications variable documented in Section III-D. This departure from normality does not undermine the validity of the fixed effects estimator itself, which does not require normally distributed errors to produce a consistent estimate of the coefficient of interest. It does, however, reinforce the importance of relying on clustered standard errors rather than the classical standard errors that underlie the conventional F-statistic discussed above, since inference based on an assumption of well-behaved, independently distributed errors would be poorly suited to data with this residual structure.

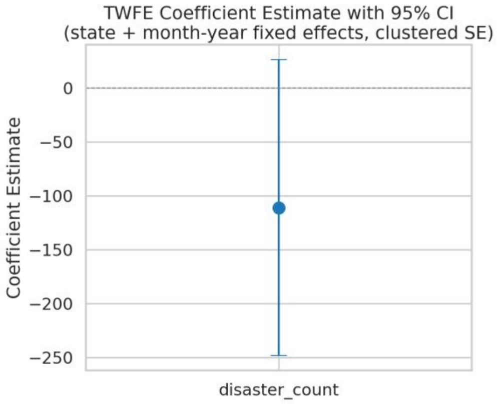

Fig. 5. TWFE coefficient estimate with 95% confidence interval.

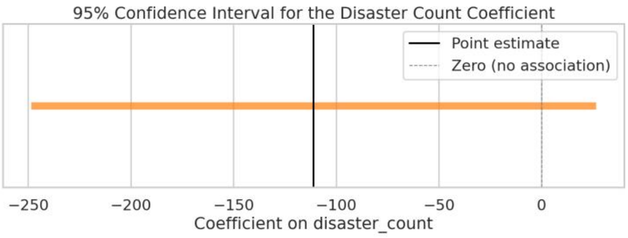

Fig. 6. 95% confidence interval, range representation.

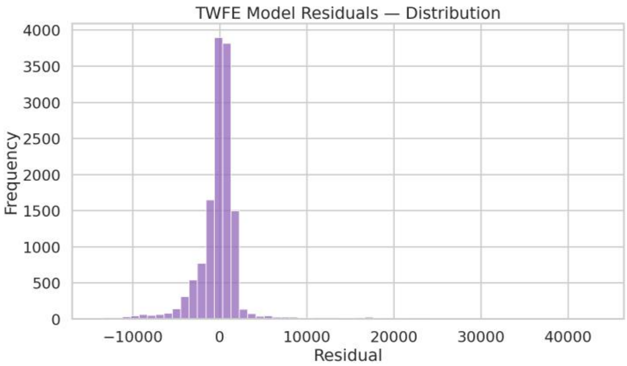

Fig. 7. Distribution of model residuals.

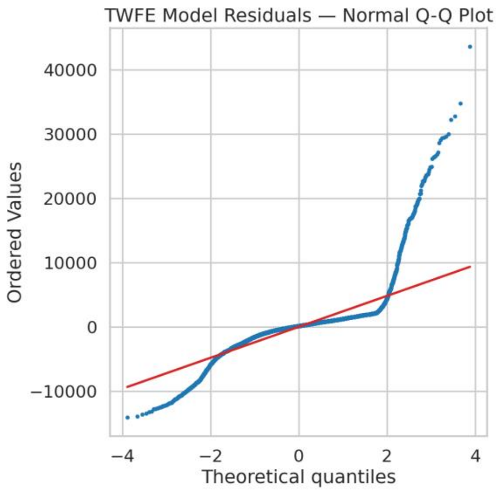

Fig. 8. Normal quantile-quantile plot of model residuals.

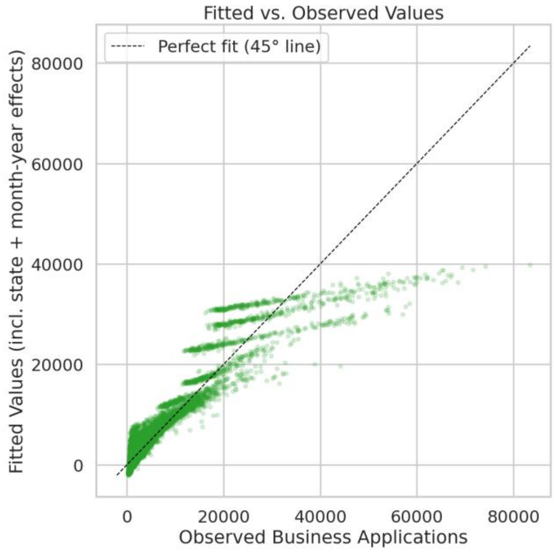

Fig. 9. Fitted values (including state and month-year effects) vs. observed business applications.

Fig. 9 presents fitted values against observed business applications, which shows the same limited explanatory power reflected in the low R^2 values reported above. Taken together, the diagnostics support the inferential approach adopted in this study, without providing grounds for stronger claims than those already advanced in Section V-B.

D. Robustness Checks

One robustness check was carried out in the course of this study, and it is important to be precise about its scope. The coefficient reported in Section V-B was independently reproduced using a separate estimation approach: rather than relying on the absorbed-effects implementation provided by the `linearmodels` package, the same regression was estimated as an explicit dummy-variable ordinary least squares model using the `statsmodels` package [30], with state and month-year indicator variables included directly rather than absorbed. Both implementations returned the same coefficient on disaster count to a high degree of numerical precision, differing only at a level consistent with floating-point rounding rather than any substantive discrepancy.

This check should be understood strictly as a verification of computational correctness — confirmation that the reported estimate is not an artifact of how a particular software package absorbs fixed effects — rather than as a robustness analysis of the underlying empirical specification. It provides confidence that the number reported in Section V-B is what the stated model actually implies, given the data, but it says nothing about whether that model's assumptions or functional form are themselves well suited to the underlying question.

Several forms of robustness analysis that would speak to the latter concern were not undertaken in this study. Standard errors were not re-estimated under alternative clustering strategies, such as two-way clustering by state and month-year; the model does not include lagged disaster counts, and so cannot speak to whether an association emerges with a delay rather than within the same month; no alternative functional forms were estimated, such as a logarithmic or inverse hyperbolic sine transformation of business applications, which might be better suited to its right-skewed distribution; disaster counts were not weighted or replaced by a measure of disaster severity; and no sub-sample analysis was conducted to assess whether the result is sensitive to unusual periods within the sample, such as the disruption associated with the COVID-19 pandemic. Each of these represents a natural and valuable extension of the present analysis rather than a check that has already been performed, and each is revisited as a direction for future research later in this paper. The absence of these additional checks should be read as a limitation on how much confidence can be placed in the precise magnitude and functional form of the estimated relationship, not as evidence that the relationship has been more thoroughly interrogated than is actually the case.

VI. DISCUSSION

A. Interpretation of the Estimated Association

The central empirical finding of this study is that, once persistent differences across states and nationwide monthly shocks are accounted for, there is no strong evidence of a systematic short-run association between federally declared disasters and state-level business applications. This finding warrants careful interpretation, because it would be a mistake to read it as evidence that disasters carry no economic consequences for entrepreneurial activity, or that the question motivating this study was misplaced. The absence of a precisely estimated relationship in a state-month panel is a considerably narrower finding than a claim that disasters do not matter for business formation, and the gap between these two statements is the proper subject of this discussion.

Entrepreneurial responses to disaster events are plausibly heterogeneous along several dimensions that a single monthly disaster count, aggregated to the state level, cannot capture. The severity of a disaster is likely to matter a great deal: a localized flood affecting a handful of counties is treated identically in this analysis to a hurricane causing widespread destruction across a state, since both register as a single declaration in the month in which they begin. Local institutional capacity — the speed and effectiveness with which state and municipal governments coordinate relief and recovery efforts — is also likely to shape whether disruption or reconstruction dominates a local economy's short-run trajectory, yet this capacity is not something the disaster count variable can distinguish across states. Similarly, entrepreneurs' access to finance and insurance coverage in the affected area plausibly determines whether a disaster becomes an opportunity for rebuilding-driven business formation or a period of prolonged disruption during which new ventures are deferred. Demographic characteristics of the affected population, the industrial composition of the local economy — particularly the presence of construction, contracting, and other recovery-adjacent sectors — and the specific design of federal and state recovery policies each plausibly condition how a disaster translates into entrepreneurial activity, as does the timing of reconstruction relative to the disaster itself, which may generate effects that are not confined to the same calendar month as the initiating event.

A state-month disaster count is, at best, a coarse proxy for this underlying bundle of mechanisms. It records that a disaster was declared, but it is silent on nearly everything that determines whether that declaration corresponds to a modest, contained disruption or a substantial regional economic shock, and it is similarly silent on the institutional and financial context within which any entrepreneurial response would occur. Given this, a null or imprecise result at this level of aggregation is not surprising on its own terms; it is consistent with a setting in which the true relationship between disasters and business formation depends heavily on factors this study's data cannot observe, rather than with a setting in which no such relationship could plausibly exist.

B. Reconciling the Descriptive and Fixed-Effects Results

One of the more informative aspects of this study is not the final coefficient itself, but the fact that the raw descriptive correlation and the fixed-effects estimate point in different directions. The descriptive statistics in Section V-A showed a weak positive correlation between disaster counts and business applications; the fixed-effects regression in Section V-B produced a negative, statistically insignificant coefficient. Reconciling these two results is useful precisely because it illustrates a general methodological point that extends well beyond this particular application.

The raw correlation is computed across all of the variation present in the panel, which includes both differences between states and differences within a state over time. Large states — California and Texas prominent among them, as shown in the descriptive analysis of FEMA declarations — generate substantially more business applications than smaller states simply as a function of their population and economic scale, and they also accumulate more federally declared disasters over the sample period, simply as a function of their geographic size and exposure to a wider range of hazard types. Because both quantities happen to be larger in the same set of large states, a raw correlation computed across all states will pick up part of this shared association with state size, even if no meaningful relationship exists between disaster timing and business formation within any individual state.

The fixed-effects specification removes exactly this source of association by allowing each state to have its own average level of business applications and its own average disaster frequency, and asking only whether deviations from a state's own typical pattern in disaster activity coincide with deviations from that state's own typical pattern in business applications, in the same month. Once this between-state variation is removed, much of the apparent positive relationship disappears, and what remains is a comparatively weak, imprecisely estimated, and slightly negative within-state association. This does not mean that the fixed-effects estimate is the "correct" relationship in some absolute sense while the raw correlation is simply wrong; rather, the two analyses answer different empirical questions. The raw correlation describes how disaster-prone states compare to disaster-light states in their overall level of business formation. The fixed-effects estimate describes how a given state's business formation moves, within the same month, when that state's own disaster activity is unusually high or low relative to its typical pattern. Confusing these two questions — treating a between-state comparison as though it answered a within-state question — is precisely the kind of identification problem that a two-way fixed effects design is intended to guard against, and this study's results offer a concrete illustration of why that distinction matters for empirical work in entrepreneurship research, where cross-sectional comparisons across regions are common and can be misleading if not carefully separated from within-region variation over time.

C. Why the Findings Should Not Be Interpreted as Evidence of No Relationship

It is important to distinguish between two claims that are easily conflated: an absence of evidence for a relationship, and evidence that no relationship exists. This study supports only the former. The confidence interval surrounding the disaster count coefficient is wide, spanning both economically meaningful negative values and economically meaningful positive values, in addition to zero. A confidence interval of this width is compatible with a genuine decline in business applications following a disaster, a genuine increase associated with post-disaster rebuilding, or an effect close to zero — and the data underlying this analysis do not permit a confident choice among these possibilities. Concluding from this evidence that disasters have no bearing on entrepreneurial activity would be a considerably stronger claim than the analysis can support; the honest conclusion is narrower — that this study's specification, applied to this particular panel, does not detect a relationship with sufficient precision to distinguish it from zero.

There are several plausible reasons why a true relationship, if one exists, might be difficult to detect using a simple state-month specification of this kind. Entrepreneurial responses to disaster may unfold with a delay rather than within the same calendar month as the triggering event: business formation tied to reconstruction activity, insurance settlements, or federal recovery funding plausibly takes hold only after some lag, a pattern this study's contemporaneous specification is not designed to capture. It is also possible that opposing mechanisms operate simultaneously and partially offset one another within the same month — disruption suppressing some new business activity at the same time that early recovery spending stimulates other new business activity — in which case a single monthly coefficient would understate the magnitude of either effect considered separately. Measurement limitations may also play a role: a federal disaster declaration is an administrative and political outcome as much as a physical one, and it may not align precisely with the timing or intensity of the underlying economic disruption. Finally, substantial heterogeneity across disaster types — a wildfire, a hurricane, and a severe winter storm plausibly having very different implications for local business formation — is collapsed into a single count in this analysis, which may obscure relationships that exist for some hazard types but not others.

Taken together, these considerations suggest that the appropriate response to this study's results is not to treat the question as settled, but to treat the current specification as a first, necessarily coarse pass at a relationship that likely requires more refined tools to detect. The findings reported here are best understood as an invitation to more targeted future research — incorporating disaster severity, allowing for lagged effects, and disaggregating by hazard type or industry — rather than as a final word on whether, or how, disasters shape entrepreneurial activity in the United States.

VII. REPRODUCIBILITY

Reproducibility is a central concern for empirical research of this kind, particularly when a study combines two independently maintained public datasets, applies a nontrivial sequence of cleaning and aggregation decisions to each, and then estimates a statistical model whose validity depends on those upstream decisions. Every descriptive statistic, figure, and regression result presented in this paper is generated through a documented computational workflow rather than through manual data manipulation, and the corresponding code is organized so that each step of that workflow can be inspected and re-executed independently.

The accompanying repository [GitHub Repository] follows a modular directory structure intended to separate distinct concerns of the project. The `src/` directory contains the production analysis code, organized into the sequential modules described below. The `config/` directory holds the project's configuration file, which records the specific filtering choices, variable names, and file paths used throughout the pipeline, so that key methodological decisions — such as which BFS series to use or which FEMA fields to retain — are declared explicitly rather than embedded implicitly within the code. The `data/` directory defines the expected locations for raw source files and for the processed panel produced by the pipeline. The `outputs/` directory holds the figures, tables, and text reports generated by the analysis. The `tests/` directory contains the automated test suite described below, and `notebooks/` provides space for exploratory work conducted outside the main pipeline. The `streamlit_app/`

directory contains the interactive dashboard discussed later in this section.

The analytical pipeline itself is organized as four sequentially dependent modules, each responsible for one clearly defined stage of the workflow and each independently runnable from the command line. The first module, `data_loader.py`, loads the two raw source files and performs structural validation to confirm that they can be read correctly, without applying any cleaning or transformation. The second module, `preprocessing.py`, performs the cleaning, filtering, and aggregation steps described in Section III — resolving BFS disclosure codes, collapsing FEMA's county-level records to distinct state-level disaster events, and merging the two sources into the state-month analysis panel. The third module, `eda.py`, generates the descriptive statistics, tables, and figures presented in Section V-A, without fitting any statistical model. The fourth module, `modeling.py`, implements the two-way fixed effects regression described in Section IV, and produces the regression tables, diagnostic figures, and the plain-language summary of results referenced throughout Section V. Running these four modules in sequence reproduces the entire empirical analysis reported in this paper from the original public source files.

In addition to this pipeline, the repository includes a companion interactive dashboard Streamlit Dashboard, organized into five pages. The Project Overview page summarizes the research question, data sources, and headline result. The Dataset Explorer page allows a reader to inspect the constructed analysis panel directly, including its dimensions, a sample of underlying rows, and summary statistics broken down by state. The Exploratory Data Analysis page reproduces the descriptive comparisons discussed in Section V-A in an interactive form, allowing a reader to filter by state and by time period. The Regression Results page presents the fixed effects regression output alongside a plain-language explanation of the coefficient, standard error, and confidence interval intended for a reader without a statistical background. The Conclusions page summarizes the main findings and limitations discussed in this paper. This dashboard is intended as a supplementary tool for exploring the underlying data and results interactively; it does not extend the analysis beyond what is reported in this manuscript, and the figures and tables presented here remain the authoritative record of the study's findings.

The analysis is implemented entirely in Python, with dependencies recorded in the repository's `requirements.txt` file. The principal packages used are `pandas` [31] and `numpy` [32] for data manipulation, `linearmodels` [29] for estimation of the two-way fixed effects model, `statsmodels` [30] for the independent verification described below, `matplotlib` [33] and `seaborn` [34] for static figure generation, and `streamlit` [35] for the interactive dashboard described above.

The repository includes an automated test suite of approximately 54 tests covering the data loading, preprocessing, exploratory analysis, and modeling modules, intended to verify that each stage of the pipeline behaves as expected on both synthetic and real data. In addition to this test suite, the regression coefficient reported in Section V-B was independently reproduced using a second, structurally distinct implementation: an explicit dummy-variable ordinary least squares model estimated with `statsmodels`, rather than the absorbed fixed-effects approach used by `linearmodels`. As discussed in Section V-D, the two implementations returned the same coefficient to a high degree of numerical precision. This cross-implementation check should be understood as a verification of computational correctness — confirmation that the reported estimate follows correctly from the stated model given the data — rather than as an econometric robustness check of the underlying specification, and it does not bear on the causal interpretability of the result.

Both datasets underlying this study originate from publicly available U.S. government sources: the Business Formation Statistics program of the U.S. Census Bureau [27], and the Disaster Declarations Summaries dataset published through FEMA's OpenFEMA platform [28]. Neither the raw source files nor the processed analysis panel are redistributed within the repository itself; both are excluded from version control, in keeping with standard practice for datasets that are large, that originate from external sources subject to their own update cycles, and that can be exactly reconstructed from those sources. What the repository does provide is the complete preprocessing pipeline necessary to reconstruct the analytical dataset from the original public files, so that a reader who obtains the two source datasets from the Census Bureau and FEMA directly can regenerate every processed file, table, and figure referenced in this paper.

Taken together, a documented and modular computational pipeline, an automated test suite, an independent cross-implementation check of the central regression estimate, and reliance exclusively on publicly available source data are intended to make the empirical claims of this paper as transparent and independently verifiable as possible, and to allow other researchers to extend, adapt, or challenge this analysis in future work with a clear record of exactly how the reported results were obtained.

VIII. LIMITATIONS

The findings reported in this study should be interpreted within the boundaries of several methodological choices made in its design. These choices were deliberate and appropriate for the research question as posed, but they also define the scope within which the results can reasonably be generalized, and they point toward a number of natural extensions for future work.

The first limitation concerns the nature of the independent variable itself. This study measures disaster occurrence through federally declared disasters recorded by FEMA, rather than through a direct physical measure of hazard intensity such as wind speed, flood extent, or estimated property damage. A federal disaster declaration is an administrative outcome, shaped by eligibility criteria, state-level requests, and a federal review process, in addition to the underlying physical event. Two events of similar physical severity may therefore be declared differently depending on state capacity, the timing of the request, or the specific thresholds applied during federal review, while, conversely, a formal declaration does not guarantee a uniform degree of economic disruption across the affected area. As a result, the disaster count variable used in this analysis should be understood as a proxy for federally recognized disaster activity rather than a precise measure of the underlying economic shock experienced by a given state in a given month.

A second limitation concerns the level of spatial aggregation at which the analysis is conducted. The empirical model treats each state as a single unit, yet most disasters affect only a subset of a state's counties or metropolitan areas rather than the state as a whole. Aggregating to the state level necessarily discards the more localized variation in economic activity that a county-level analysis could capture, and it treats a disaster confined to a small portion of a large state in the same way as one affecting a comparable share of a smaller state. A more geographically granular analysis, matching disaster-affected counties directly to local measures of entrepreneurial activity, would likely be better positioned to detect economic responses that are concentrated in the immediate vicinity of a disaster and that may be diluted beyond recognition once aggregated to the state level.

A third limitation concerns the temporal structure of the empirical specification. The model estimated in this study captures only the contemporaneous association between a disaster declared in a given month and business applications filed in that same month. Entrepreneurial responses to disaster, however, plausibly unfold over a longer horizon: business formation tied to reconstruction activity, insurance settlements, or the securing of financing is likely to lag the disaster itself by weeks or months, as displaced business owners and prospective entrepreneurs assess damage, arrange capital, and plan new ventures. A specification incorporating lagged disaster measures would be better suited to detecting effects of this kind, and represents a natural next step building on the present analysis, though it falls outside the scope of the current study.

A related limitation concerns the treatment of disasters as a simple count, with no distinction made among events of differing severity, geographic extent, or economic consequence. A minor, localized event and a catastrophic, wide-reaching one are recorded identically in the disaster count variable used here, provided each corresponds to a distinct federal declaration. This treatment simplifies the empirical design considerably but almost certainly obscures heterogeneity in entrepreneurial response that a severity-weighted measure, or a measure incorporating direct estimates of economic loss, would be better equipped to capture.

The model specification itself is intentionally parsimonious. Business applications are regressed on disaster count alone, together with state and month-year fixed effects, rather than on a fuller set of time-varying covariates such as state unemployment rates, income growth, credit market conditions, demographic shifts, or state-level policy variables. This choice reflects the study's focus on a specific and clearly defined research question — the within-state association between disaster timing and business formation, net of persistent state characteristics and common national shocks — rather than an attempt to build a comprehensive predictive model of business formation. The omission of these additional covariates does not undermine the validity of the reported association as an answer to the question the model is designed to address, but a richer specification incorporating such covariates could help clarify whether the estimated relationship is sensitive to other time-varying local economic conditions.

Finally, and most fundamentally, this remains an observational study. The two-way fixed effects design substantially reduces the scope for confounding by removing time-invariant differences across states and shocks common to all states in a given month, but it does not, and cannot, establish a causal relationship between disasters and business formation. State-specific factors that vary over time and happen to coincide with both disaster timing and shifts in entrepreneurial activity are not ruled out by this design, nor is the possibility that unobserved local conditions jointly influence both variables. Future research seeking a more causally interpretable estimate might consider event-study designs that trace the trajectory of business formation before and after a disaster in greater temporal detail, difference-in-differences approaches that exploit comparisons between similarly situated states that did and did not experience a given disaster, or instrumental variable strategies that isolate variation in disaster exposure attributable to purely physical or meteorological factors. These approaches should be viewed as complementary to the present analysis rather than as strictly superior alternatives, each carrying its own identifying assumptions and data requirements that would need to be carefully justified in their own right.

Considered together, these limitations do not so much weaken the present findings as clarify the terms on which they should be read: as evidence concerning a specific, contemporaneous, state-level association, estimated under a deliberately focused and transparent specification. They also constitute a fairly direct research agenda, identifying several concrete directions — finer spatial resolution, lagged and severity-weighted disaster measures, richer covariate sets, and more explicitly causal identification strategies —

through which future work could build on the approach developed here to arrive at a more complete understanding of how disasters shape entrepreneurial activity in the United States.

IX. CONCLUSION AND FUTURE WORK

This study set out to answer a straightforward but empirically demanding question: does the occurrence of a federally declared disaster in a U.S. state coincide with a change in that state's monthly rate of new business formation, once persistent differences across states and shared nationwide conditions are taken into account. The question sits at the intersection of entrepreneurship research and regional economics, and it carries practical relevance for policymakers and local officials who must decide how to allocate recovery resources after a disaster, and for researchers interested in the resilience of entrepreneurial activity in the face of environmental shocks. Despite growing interest in the economic consequences of natural disasters, relatively little systematic evidence exists on their short-run relationship with new business formation specifically, as distinct from broader measures of economic output or employment, and this study was designed to address that gap using a transparent and reproducible empirical approach.

To do so, this paper constructed a state-month panel spanning July 2004 through the most recent available data, combining Business Formation Statistics from the U.S. Census Bureau with Disaster Declarations Summaries from FEMA. The panel covers 51 units of geography — the fifty states and the District of Columbia — over 264 months in the estimation sample, yielding 13,464 state-month observations. The empirical strategy centered on a two-way fixed effects regression of business applications on disaster count, with state fixed effects absorbing time-invariant differences across states and month-year fixed effects absorbing shocks common to all states in a given month, with standard errors clustered at the state level to account for serial correlation in state-level disturbances over time.

The principal empirical finding is that the estimated association between disaster declarations and business applications, under this specification, is not statistically distinguishable from zero at conventional significance levels. The point estimate is negative, but the associated confidence interval is wide and includes both economically meaningful positive and negative values. This result stands in some contrast to the weak positive correlation apparent in the raw, unadjusted data, and Section VI argued that this divergence is itself informative: it illustrates how a raw cross-state correlation can reflect the fact that larger, more disaster-exposed states also tend to generate more business applications overall, rather than any genuine within-state relationship between disaster timing and business formation. Once this between-state variation is removed by the fixed effects design, the apparent association weakens considerably and loses statistical precision. This finding should not be read as demonstrating that disasters carry no consequences for entrepreneurial activity; rather, it indicates that a simple contemporaneous, state-month specification, using an administrative count of federal disaster declarations, does not detect a relationship with sufficient statistical power to draw firm conclusions in either direction.

This result contributes to the entrepreneurship and regional economics literature in a modest but genuine way. Much of the existing work on disasters and local economies has focused on aggregate output, employment, or population movements following major events [5], [6], with comparatively less attention paid specifically to the rate at which new businesses are formed in the aftermath of disaster. By focusing squarely on business applications as the outcome of interest, and by applying a design that explicitly separates within-state from between-state variation, this study offers a methodologically disciplined benchmark against which more targeted future analyses can be compared. Perhaps more importantly, this paper also contributes a fully transparent and reproducible empirical workflow: a documented pipeline linking two independent federal data sources into an analysis-ready panel, an open and modular codebase supporting each stage of data cleaning, exploration, and estimation, an accompanying interactive dashboard that allows a reader to inspect the underlying data and regression output directly, and an automated test suite verifying the correctness of each analytical step. Combined with an independent cross-implementation check confirming that the reported coefficient is not an artifact of a particular software package, these elements are intended to make the empirical claims of this paper unusually easy to verify, scrutinize, and extend.

It is worth commenting briefly on the scientific value of reporting a statistically non-significant result of this kind rather than continuing to search for a specification that yields significance. A published literature that systematically reports only significant findings risks overstating the strength and consistency of empirical relationships in a given domain, a concern that has received considerable attention across the social sciences [36]. Transparently reporting that a reasonable, well-motivated specification does not detect a significant association — while being explicit about the specification's limitations and about what would be required to detect a more precisely estimated effect — provides a more honest contribution to cumulative knowledge than a result obtained by iterating over specifications until statistical significance is achieved. In this sense, the absence of a significant finding here is not merely a null outcome to be set aside, but a substantive piece of evidence about the difficulty of detecting this relationship using administrative disaster counts at the state-month level, one that can usefully inform the design of future studies on this topic.

Several extensions follow naturally from the limitations discussed in Section VIII and would meaningfully advance understanding of this relationship. Incorporating lagged disaster measures would allow the analysis to capture entrepreneurial responses that unfold over several months rather than within the same period as the triggering event, which is plausible given that reconstruction activity, insurance settlements, and new business financing typically take time to materialize. Replacing the simple disaster count with a severity-weighted measure, drawing on estimated damage costs or declaration type, would allow the analysis to distinguish the economic consequences of catastrophic events from those of comparatively minor, contained ones, which the present specification cannot do. Moving to a county-level or metropolitan-area unit of analysis would recover the substantial within-state geographic variation that is necessarily lost when aggregating to the state level, and would likely be better suited to detecting effects concentrated in the immediate vicinity of a disaster. Disaggregating business applications by industry sector would allow future work to test whether construction, contracting, and other recovery-adjacent industries respond differently from the broader economy, a distinction the current state-level total cannot reveal. Extending the model to include richer time-varying covariates, such as state unemployment rates, income growth, or credit market conditions, would help clarify whether the estimated relationship is sensitive to other local economic forces evolving alongside disaster activity. An event-study design tracing business formation in the months immediately before and after a disaster would provide a more detailed picture of the dynamics involved than a single pooled coefficient can offer, while difference-in-differences approaches comparing similarly situated states that did and did not experience a comparable disaster could sharpen the comparison further, provided a credible set of comparison states can be identified. Instrumental variable strategies built on purely physical hazard measures — wind speed, rainfall totals, or seismic intensity, for instance — rather than administrative declarations, would move the analysis closer to a causally interpretable estimate by addressing the concern, raised in Section VIII, that federal declarations reflect an administrative process in addition to the underlying physical event. Extending the sample to a longer time horizon as additional years of data become available would also improve the statistical power available to detect a modest effect, an important consideration given the width of the confidence interval reported in this study. Finally, additional robustness analyses — alternative clustering strategies, alternative functional forms better suited to the right-skewed distribution of business applications, and sub-sample analyses excluding unusual periods such as the COVID-19 pandemic — would help establish whether the reported estimate is sensitive to specification choices that this study did not explore.

Each of these extensions addresses a distinct source of imprecision or potential bias in the present analysis, and pursuing several of them in combination is likely to be more informative than any one in isolation. Taken together, they define a coherent research agenda building directly on the panel, pipeline, and empirical framework developed here, rather than requiring an entirely new approach to the question.

More broadly, this study is offered as one example of how questions in entrepreneurship and disaster economics can be pursued using publicly available federal data and fully reproducible computational methods, without overstating what a single, focused specification can establish. The relationship between disaster exposure and entrepreneurial activity in the United States remains an open and practically important question, and the transparent reporting of a carefully specified, honestly interpreted result — including one that does not reach conventional statistical significance — is intended as a useful and durable contribution to that ongoing inquiry, and as a foundation others can build upon with the more targeted extensions outlined above.

DATA AVAILABILITY STATEMENT

Both datasets analyzed in this study are publicly available from their originating U.S. federal agencies: Business Formation Statistics from the U.S. Census Bureau (https://www.census.gov/econ/bfs/) and Disaster Declarations Summaries from FEMA's OpenFEMA platform (https://www.fema.gov/openfema-data-page/disaster-declarations-summaries-v2). Neither the raw source files nor the processed analysis panel are redistributed within the accompanying repository; both can be exactly reconstructed from the original public sources using the preprocessing pipeline described in Section VII.

CODE AVAILABILITY STATEMENT

All code used to produce the results in this paper — data loading, preprocessing, exploratory analysis, and regression modeling — is openly available at [GitHub Repository](). The repository includes a configuration file recording all filtering and modeling parameters, an automated test suite, and an interactive Streamlit dashboard: [Live App]() for exploring the underlying data and results.

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CONFLICT OF INTEREST STATEMENT

The author declares no competing interests.

AUTHOR CONTRIBUTIONS

Anupriyo Chakraborty: Conceptualization, Methodology, Software, Data Curation, Formal Analysis, Validation, Visualization, Writing – Original Draft, Writing – Review & Editing, Project Administration.

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